THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Sides 15 cm, 9 cm and 5 cm are given. Which statement is true? 1

(a) only an equilateral triangle is possible **(b)** a triangle can be formed
(c) only a right triangle is possible **(d)** no triangle can be formed

2. In an isosceles triangle with perimeter 41 cm the equal sides are 15 cm each. The base (cm) is: 1

(a) 12 **(b)** 13 **(c)** 10 **(d)** 11

3. The two acute angles of a right triangle are in the ratio $4 : 5$. The smaller acute angle is: 1

(a) 60° **(b)** 50° **(c)** 30° **(d)** 40°

4. The number of isosceles triangles with whole-number sides and perimeter 18 cm (equilateral included) is: 1

(a) 4 **(b)** 6 **(c)** 3 **(d)** 5

5. The number of triangles with whole-number sides (in cm) and perimeter 15 cm is: 1

(a) 6 **(b)** 7 **(c)** 9 **(d)** 8

6. The exterior angle at the base of an isosceles triangle is 150°. The apex angle is: 1

(a) 140° **(b)** 110° **(c)** 130° **(d)** 120°

7. A triangle cannot have: 1

(a) two right angles **(b)** two acute angles
(c) one obtuse angle **(d)** three acute angles

8. The angles of a triangle are $(5x)^\circ$, $(2x + 4)^\circ$ and $(4x - 44)^\circ$. The largest angle is: 1

(a) 90° **(b)** 100° **(c)** 120° **(d)** 110°

9. **Assertion (A):** Two angles of a triangle are 50° and 60°. The third angle is 70°. 1
Reason (R): The angle sum of a triangle is 180°.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Sides of 5 cm, 7 cm and 10 cm can form a triangle. 1
Reason (R): In a triangle the sum of any two sides is greater than the third side.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. List all isosceles triangles with whole-number sides and perimeter 24 cm (equal sides a, base b). How many are there? 2

12. In $\triangle ABC$ and $\triangle PQR$ it is known that $\angle B = \angle Q = 45^\circ$, $BC = QR = 9$ cm and $\angle C = \angle R = 80^\circ$. Are the triangles necessarily congruent? Give the rule or the reason. 2

13. In isosceles $\triangle ABC$ with $AB = AC$, side BC is extended to D and $\angle ACD = 160^\circ$. Find $\angle A$ and $\angle B$. 2

14. List all isosceles triangles with whole-number sides and perimeter 22 cm (equal sides a, base b). How many are there? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, $\angle A = 30^\circ$. The exterior angles at B and C are $(3x + 50)^\circ$ and $(2x + 60)^\circ$. Find x and the angles B and C. 3

16. The perimeter of an isosceles triangle is 52 cm. Each equal side is 5 cm longer than the base. Find the sides and show that such a triangle exists. 3

17. Two sides of a triangle are 7 cm and 25 cm. The third side is a whole number of cm divisible by 5. List all possible lengths and the largest perimeter. 3

OR

In $\triangle ABC$, $\angle A = 80^\circ$. The exterior angles at B and C are $(7x + 58)^\circ$ and $(5x + 130)^\circ$. Find x and the angles B and C. 3

18. The perimeter of an isosceles triangle is 34 cm. Each equal side is 2 cm longer than the base. Find the sides and show that such a triangle exists. 3

OR

Two sides of a triangle are 12 cm and 31 cm. The third side is a whole number of cm divisible by 7. List all possible lengths and the largest perimeter. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A gable of a house is an isosceles triangle ABC with AB = AC and apex angle $\angle A = 40^\circ$. 5

(i) Find each base angle.
(ii) Name the triangle by its angles.
(iii) Find the exterior angle at B.
(iv) The equal sides are 8 m each. Find the perimeter if the base is 5 m. Can the base be 17 m? Give a reason.

20. The exterior angles of a triangle (one at each vertex) are in the ratio $3 : 5 : 7$. Find the three exterior angles, the three interior angles and classify the triangle. 5

OR

$\triangle ABC \cong \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Given $\angle A = (3x + 14)^\circ$, $\angle D = (5x - 22)^\circ$, $AB = (2y + 6)$ cm and $DE = (3y - 4)$ cm. Find x, y, $\angle A$ and AB. If $\angle B = 50^\circ$, find $\angle F$. 5

END OF QUESTION PAPER — Total: 40 Marks